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LAB: AI-Based Telecom Customer Churn Analytics

Database: MSSQL Server

Use Case: Telecom Customer Churn Prediction

=====

1. Create Database

```
8 CREATE DATABASE TelecomAI_DB;
9 GO
10
11 USE TelecomAI_DB;
12 GO
```

=====

2. Create Customer Table

=====

```
18 CREATE TABLE telecom_customers (
19     customer_id INT PRIMARY KEY,
20     customer_name VARCHAR(100),
21     gender VARCHAR(10),
22     age INT,
23     contract_type VARCHAR(20),
24     monthly_bill DECIMAL(10,2),
25     internet_usage_gb DECIMAL(10,2),
26     call_minutes INT,
27     complaint_count INT,
28     payment_delay_days INT,
29     churn_status VARCHAR(10)
30 );
31 GO
```

=====

3. Insert Sample Telecom Data

=====

```

37  INSERT INTO telecom_customers
38  (customer_id, customer_name, gender, age, contract_type, monthly_bill,
39   internet_usage_gb, call_minutes, complaint_count, payment_delay_days, churn_status)
40  VALUES
41  (1, 'Aung Aung', 'Male', 28, 'Prepaid', 25000, 35.5, 400, 3, 10, 'Yes'),
42  (2, 'Su Su', 'Female', 32, 'Postpaid', 45000, 80.0, 900, 0, 0, 'No'),
43  (3, 'Kyaw Kyaw', 'Male', 45, 'Prepaid', 18000, 20.0, 250, 4, 15, 'Yes'),
44  (4, 'Mya Mya', 'Female', 25, 'Postpaid', 60000, 120.5, 1200, 1, 2, 'No'),
45  (5, 'Hla Hla', 'Female', 39, 'Prepaid', 22000, 25.5, 300, 5, 20, 'Yes'),
46  (6, 'Tun Tun', 'Male', 30, 'Postpaid', 52000, 100.0, 1000, 0, 0, 'No'),
47  (7, 'Nandar', 'Female', 27, 'Prepaid', 20000, 18.5, 200, 2, 8, 'Yes'),
48  (8, 'Moe Moe', 'Female', 35, 'Postpaid', 48000, 75.0, 850, 1, 1, 'No'),
49  (9, 'Zaw Zaw', 'Male', 42, 'Prepaid', 26000, 30.0, 350, 4, 12, 'Yes'),
50  (10, 'Ei Ei', 'Female', 29, 'Postpaid', 55000, 110.0, 1100, 0, 0, 'No');
51  GO

```

4. Data Exploration

View all data

58 **SELECT * FROM telecom_customers;**

100 % No issues found

Results Messages

	customer_id	customer_name	gender	age	contract_type	monthly_bill	internet_usage_gb	call_minutes	complaint_count	payment_delay_days	churn_status
1	1	Aung Aung	Male	28	Prepaid	25000.00	35.50	400	3	10	Yes
2	2	Su Su	Female	32	Postpaid	45000.00	80.00	900	0	0	No
3	3	Kyaw Kyaw	Male	45	Prepaid	18000.00	20.00	250	4	15	Yes
4	4	Mya Mya	Female	25	Postpaid	60000.00	120.50	1200	1	2	No
5	5	Hla Hla	Female	39	Prepaid	22000.00	25.50	300	5	20	Yes
6	6	Tun Tun	Male	30	Postpaid	52000.00	100.00	1000	0	0	No
7	7	Nandar	Female	27	Prepaid	20000.00	18.50	200	2	8	Yes
8	8	Moe Moe	Female	35	Postpaid	48000.00	75.00	850	1	1	No
9	9	Zaw Zaw	Male	42	Prepaid	26000.00	30.00	350	4	12	Yes
10	10	Ei Ei	Female	29	Postpaid	55000.00	110.00	1100	0	0	No

Count churn and non-churn customers

68 **SELECT**

69 **churn_status,**

70 **COUNT(*) AS total_customers**

71 **FROM telecom_customers**

72 **GROUP BY churn_status;**

100 % No issues found

Results Messages

	churn_status	total_customers
1	No	5
2	Yes	5

Average monthly bill by churn status

```
79  SELECT
80      churn_status,
81      AVG(monthly_bill) AS avg_monthly_bill
82  FROM telecom_customers
83  GROUP BY churn_status;
84
```

100 % No issues found

	churn_status	avg_monthly_bill
1	No	52000.000000
2	Yes	22200.000000

Average complaints by churn status

```
86  SELECT
87      churn_status,
88      AVG(complaint_count) AS avg_complaints
89  FROM telecom_customers
90  GROUP BY churn_status;
```

100 % No issues found

	churn_status	avg_complaints
1	No	0
2	Yes	3

=====

5. Data Preparation

Convert Churn Status into Numeric Label

Yes = 1, No = 0

=====

103	SELECT
104	customer_id,
105	customer_name,
106	age,
107	contract_type,
108	monthly_bill,
109	internet_usage_gb,
110	call_minutes,
111	complaint_count,
112	payment_delay_days,
113	CASE
114	WHEN churn_status = 'Yes' THEN 1
115	ELSE 0
116	END AS churn_label
117	FROM telecom_customers;

100 %
✔ No issues found

	customer_id	customer_name	age	contract_type	monthly_bill	internet_usage_gb	call_minutes	complaint_count	payment_delay_days	churn_label
1	1	Aung Aung	28	Prepaid	25000.00	35.50	400	3	10	1
2	2	Su Su	32	Postpaid	45000.00	80.00	900	0	0	0
3	3	Kyaw Kyaw	45	Prepaid	18000.00	20.00	250	4	15	1
4	4	Mya Mya	25	Postpaid	60000.00	120.50	1200	1	2	0
5	5	Hla Hla	39	Prepaid	22000.00	25.50	300	5	20	1
6	6	Tun Tun	30	Postpaid	52000.00	100.00	1000	0	0	0
7	7	Nandar	27	Prepaid	20000.00	18.50	200	2	8	1
8	8	Moe Moe	35	Postpaid	48000.00	75.00	850	1	1	0
9	9	Zaw Zaw	42	Prepaid	26000.00	30.00	350	4	12	1
10	10	Ei Ei	29	Postpaid	55000.00	110.00	1100	0	0	0

=====

6. Simple AI Rule-Based Churn Risk Scoring

This simulates AI scoring logic inside SQL.

=====

```

135     SELECT
136         customer_id,
137         customer_name,
138         contract_type,
139         monthly_bill,
140         complaint_count,
141         payment_delay_days,
142         churn_status,
143
144     CASE
145         WHEN complaint_count >= 4
146             OR payment_delay_days >= 15
147             OR contract_type = 'Prepaid'
148         THEN 'High Risk'
149
150         WHEN complaint_count BETWEEN 2 AND 3
151             OR payment_delay_days BETWEEN 7 AND 14
152         THEN 'Medium Risk'
153
154         ELSE 'Low Risk'
155     END AS predicted_churn_risk
156
157 FROM telecom_customers;
158 GO

```

	customer_id	customer_name	contract_type	monthly_bill	complaint_count	payment_delay_days	churn_status	predicted_churn_risk
1	1	Aung Aung	Prepaid	25000.00	3	10	Yes	High Risk
2	2	Su Su	Postpaid	45000.00	0	0	No	Low Risk
3	3	Kyaw Kyaw	Prepaid	18000.00	4	15	Yes	High Risk
4	4	Mya Mya	Postpaid	60000.00	1	2	No	Low Risk
5	5	Hla Hla	Prepaid	22000.00	5	20	Yes	High Risk
6	6	Tun Tun	Postpaid	52000.00	0	0	No	Low Risk
7	7	Nandar	Prepaid	20000.00	2	8	Yes	High Risk
8	8	Moe Moe	Postpaid	48000.00	1	1	No	Low Risk
9	9	Zaw Zaw	Prepaid	26000.00	4	12	Yes	High Risk
10	10	Ei Ei	Postpaid	55000.00	0	0	No	Low Risk

=====

7. Create View for Churn Risk Dashboard

=====

```

164 CREATE VIEW vw_customer_churn_risk AS
165 SELECT
166     customer_id,
167     customer_name,
168     gender,
169     age,
170     contract_type,
171     monthly_bill,
172     internet_usage_gb,
173     call_minutes,
174     complaint_count,
175     payment_delay_days,
176     churn_status,
177
178     CASE
179         WHEN complaint_count >= 4
180             OR payment_delay_days >= 15
181             OR contract_type = 'Prepaid'
182         THEN 'High Risk'
183
184         WHEN complaint_count BETWEEN 2 AND 3
185             OR payment_delay_days BETWEEN 7 AND 14
186         THEN 'Medium Risk'
187
188         ELSE 'Low Risk'
189     END AS predicted_churn_risk
190 FROM telecom_customers;
191 GO

```

8. Query the Churn Risk Dashboard View

```

196
197 SELECT * FROM vw_customer_churn_risk;
198

```

100 % No issues found

	customer_id	customer_name	gender	age	contract_type	monthly_bill	internet_usage_gb	call_minutes	complaint_count	payment_delay_days	churn_status	predicted_churn_risk
1	1	Aung Aung	Male	28	Prepaid	25000.00	35.50	400	3	10	Yes	High Risk
2	2	Su Su	Female	32	Postpaid	45000.00	80.00	900	0	0	No	Low Risk
3	3	Kyaw Kyaw	Male	45	Prepaid	18000.00	20.00	250	4	15	Yes	High Risk
4	4	Mya Mya	Female	25	Postpaid	60000.00	120.50	1200	1	2	No	Low Risk
5	5	Hla Hla	Female	39	Prepaid	22000.00	25.50	300	5	20	Yes	High Risk
6	6	Tun Tun	Male	30	Postpaid	52000.00	100.00	1000	0	0	No	Low Risk
7	7	Nandar	Female	27	Prepaid	20000.00	18.50	200	2	8	Yes	High Risk
8	8	Moe Moe	Female	35	Postpaid	48000.00	75.00	850	1	1	No	Low Risk
9	9	Zaw Zaw	Male	42	Prepaid	26000.00	30.00	350	4	12	Yes	High Risk
10	10	Ei Ei	Female	29	Postpaid	55000.00	110.00	1100	0	0	No	Low Risk

Show only high-risk customers

```
200 SELECT *
201 FROM vw_customer_churn_risk
202 WHERE predicted_churn_risk = 'High Risk';
203
```

100 % No issues found

	customer_id	customer_name	gender	age	contract_type	monthly_bill	internet_usage_gb	call_minutes	complaint_count	payment_delay_days	churn_status	predicted_churn_risk
1	1	Aung Aung	Male	28	Prepaid	25000.00	35.50	400	3	10	Yes	High Risk
2	3	Kyaw Kyaw	Male	45	Prepaid	18000.00	20.00	250	4	15	Yes	High Risk
3	5	Hla Hla	Female	39	Prepaid	22000.00	25.50	300	5	20	Yes	High Risk
4	7	Nandar	Female	27	Prepaid	20000.00	18.50	200	2	8	Yes	High Risk
5	9	Zaw Zaw	Male	42	Prepaid	26000.00	30.00	350	4	12	Yes	High Risk

9. Retention Campaign Recommendation

```
208 SELECT
209     customer_id,
210     customer_name,
211     predicted_churn_risk,
212     CASE
213         WHEN predicted_churn_risk = 'High Risk'
214         THEN 'Offer discount package and priority customer support'
215
216         WHEN predicted_churn_risk = 'Medium Risk'
217         THEN 'Send loyalty offer and service improvement message'
218
219         ELSE 'Maintain normal service communication'
220     END AS recommended_action
221 FROM vw_customer_churn_risk;
222 GO
```

100 % No issues found

	customer_id	customer_name	predicted_churn_risk	recommended_action
1	1	Aung Aung	High Risk	Offer discount package and priority customer sup...
2	2	Su Su	Low Risk	Maintain normal service communication
3	3	Kyaw Kyaw	High Risk	Offer discount package and priority customer sup...
4	4	Mya Mya	Low Risk	Maintain normal service communication
5	5	Hla Hla	High Risk	Offer discount package and priority customer sup...
6	6	Tun Tun	Low Risk	Maintain normal service communication
7	7	Nandar	High Risk	Offer discount package and priority customer sup...
8	8	Moe Moe	Low Risk	Maintain normal service communication
9	9	Zaw Zaw	High Risk	Offer discount package and priority customer sup...
10	10	Ei Ei	Low Risk	Maintain normal service communication

10. Model Evaluation Simulation

Compare Actual Churn Status with Predicted Risk

```
229  SELECT
230      churn_status,
231      predicted_churn_risk,
232      COUNT(*) AS total_customers
233  FROM vw_customer_churn_risk
234  GROUP BY churn_status, predicted_churn_risk;
235  GO
```

100 % No issues found

Results		Messages	
	churn_status	predicted_churn_risk	total_customers
1	Yes	High Risk	5
2	No	Low Risk	5